

Principled Design of Interpretable Automated Scoring for Large-Scale Educational Assessments

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Abstract

AI-driven automated scoring systems offer scalable and efficient means of evaluating complex student-generated responses. Yet, despite increasing demand for transparency and interpretability, the field has yet to develop a widely accepted solution for interpretable automated scoring to be used in large-scale real-world assessments. This work takes a principled approach to address this challenge. We analyze the needs and potential benefits of interpretable automated scoring for various assessment stakeholder groups and develop four principles of interpretability – Faithfulness, Groundedness, Traceability, and Interchangeability (FGTI) – targeted at those needs. To illustrate the feasibility of implementing these principles, we develop the ANALYTICSCORE framework for short answer scoring as a baseline reference framework for future research. In terms of scoring accuracy, ANALYTICSCORE outperforms many uninterpretable scoring methods and is, on average, within 0.06 QWK of the uninterpretable SOTA across 10 items from the ASAP-SAS dataset. By comparing against human annotators conducting the same featurization task, we further demonstrate that the featurization behavior of ANALYTICSCORE aligns well with that of humans.

1 Introduction

Accurate and credible assessment of knowledge and skills forms the basis for effective decision making in a variety of educational contexts, from student learning and instructional design to program development and policy making (Berman et al., 2019). When the set of knowledge and skills to be gauged involves complex, open-ended problem-solving and communication abilities, AI-driven *automated scoring systems* can offer rapid, accessible, and scalable alternatives to the otherwise labor-intensive and costly process of training and deploying human scorers (Foltz et al., 2020). Automated scoring systems have been increasingly adopted across many assessment contexts over the past several decades, achieving acceptable levels of scoring accuracy in various areas of human learning (Whitmer and Beiting-Parrish, 2023, 2024).

Despite progress, automated scoring of open-ended responses has yet to reliably obtain generalizable scoring accuracy across diverse scoring contexts. Even when automated scoring meets acceptable levels of scoring accuracy, errors or biases inherent in the scoring algorithm can profoundly harm student learning and equity, policy evaluation, and public trust (Berman et al., 2019, Pellegrino, 2022). For these reasons, improving transparency and interpretability in automated scoring has now become a moral imperative, not a mere technical preference (Holmes et al., 2022, Khosravi et al., 2022, Memarian and Doleck, 2023, Schlippe et al., 2022). Yet, in spite of the growing research on

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interpretable and explainable AI as well as its applications specifically within educational assessment, interpretable automated scoring remains mostly confined to academic research with limited adoption in large-scale, real-world assessment (Institute of Education Statistics, 2023, Whitmer and Beiting-Parrish, 2023, 2024).

In this paper, we take a principled approach towards building a practical interpretable automated scoring solution for large-scale assessments. An effective interpretability solution begins by identifying the diverse needs of each stakeholder in understanding the system’s decisions, and by grounding the development of interpretable AI systems in those needs (Bhatt et al., 2020, Páez, 2019, Preece et al., 2018). Research on interpretable automated scoring, on the other hand, has largely ignored this need-finding process. As we discuss in Section 2.2, this neglect has often led to several claimed interpretability solutions that fail to address the diverse and nuanced interpretability needs of the human actors in educational assessment.

We identify the needs and benefits of scoring model explanations for various large-scale assessment stakeholder groups consisting of test takers, assessment developers, and test users (Section 2.1). Targeted at those needs, we develop the principles of **faithful**, **grounded**, **traceable**, and **interchangeable** (FGTI) model interpretations for AI-driven automated scoring (Section 2.2).

We further illustrate the feasibility of implementing the FGTI principles in practice and establish a concrete baseline for future work (Section 3). ANALYTICSCORE is the first interpretable automated short-answer scoring framework to embody our principles. It operates by extracting explicitly identifiable elements from unannotated response texts and featurizing each response into human-interpretable values based on those elements. These features are input to an intuitive ordinal logistic regression module for scoring.

We measure the performance of ANALYTICSCORE on a real-world response dataset by measuring (1) scoring accuracy and (2) alignment of featurization behaviors with human judgments (Sections 4 and 5). ANALYTICSCORE outperforms many uninterpretable scoring methods and is within 0.06 QWK of the uninterpretable SOTA on average across 10 items from the ASAP-SAS dataset. The featurization behavior of ANALYTICSCORE also aligns well with humans (0.90, 0.72, 0.81 QWK across assessment areas). Our findings indicate strong potential for implementing accurate and well-aligned interpretability solutions that meet the real needs of assessment stakeholders.

Automated Scoring and Interpretable AI Researchers have increasingly noted the need to enhance the transparency of complex AI-driven automated scoring systems through model explanations (Bauer and Zapata-Rivera, 2020, Bennett and Zhang, 2015, Schlippe et al., 2022). Several approaches have been proposed to address this challenge, and we discuss them in detail in Section 2.2 in connection with our four principles.

Despite growing research interests, interpretable automated scoring still lacks practical adoption and meaningful field use. The 2023 NAEP Math Automated Scoring Challenge¹ for open-ended math responses organized by the US National Center for Education Statistics (NCES) found that none of the submissions met the criteria for interpretability despite several methods achieving near-human scoring accuracy (Institute of Education Statistics, 2023, Whitmer and Beiting-Parrish, 2024). Our work provides a stakeholder-centered approach towards addressing this gap.

¹<https://github.com/NAEP-AS-Challenge/math-prediction>

2 Building the Principles of Interpretable Automated Scoring

Insights derived from scoring support various human actors throughout the overall assessment process. Below we analyze three main stakeholder groups in large-scale assessment – test takers, assessment developers, and test users (AERA et al., 2014, Berman et al., 2019). Each stakeholder group’s distinct roles and priorities uniquely shape how interpretable automated scoring can improve their assessment experience.

2.1 Interpretability Needs and Benefits

Test Takers The needs and benefits of interpretable scoring vary depending on the assessment type: summative or formative. Most large-scale assessments are summative assessments, which are assessments *of* learning that support evaluating learner achievement, assigning grades, or determining proficiency levels (Harlen, 2005). Because these assessments often drive high-stakes decisions, test takers need to trust the fairness and justifiability of scoring decisions (Williamson et al., 2012). Provided that the scoring algorithm implements sound scoring logic, allowing test takers or their representatives to examine traceable explanations for scoring decisions can foster trust (Bauer and Zapata-Rivera, 2020, Ferrara and Qunbar, 2022). These explanations can also support a streamlined quality control process by facilitating the identification and correction of errors, improving the overall integrity of the assessment (see Bennett and Zhang (2015) and Ferrara and Qunbar (2022)).

Formative assessments are assessments *for* learning, intended to guide and improve learner performance through frequent practice, progress monitoring, and skill diagnosis (Black and Wiliam, 1998, Wiliam, 2011). In this context, the function of automated scoring is primarily to provide timely, effective and actionable feedback to support learner learning (Bennett, 2006, DiCerbo et al., 2020). Effective feedback should help learners understand the discrepancy between their work and a desired outcome (Schwartz et al., 2016). A step-by-step explanation of the features observed in a learner’s work, coupled with human understandable descriptions of how those features were processed can be used to provide such elaborative feedback.

Assessment Developers Scoring algorithms should reliably identify evidence of the constructs (target knowledge, skills, and abilities) measured by the task (Bejar et al., 2016). Understanding the types of evidence that an automated scoring algorithm reliably detects also informs other key aspects of assessment design, such as construct selection and task design (Bennett and Bejar, 1998). Model explanations can facilitate this understanding by transparently revealing the features used by the scoring algorithm and its intermediate reasoning steps. Explanations can also help determine which parts of the algorithm can be reused, avoiding the costly and time-consuming process of training a new scoring algorithm for each new task (see DiCerbo et al. (2020)).

Model explanations also yield specific insights into areas where the scoring model can be improved and how. Scoring models often need to be tuned for various reasons. For instance, models trained on data may reflect biases related to response strategies specific to student groups (Ferrara and Qunbar, 2022, Rupp, 2018). Scoring models may also become less stable over time as the test-taker population and/or scoring criteria change (Bejar et al., 2016). Transparent inspection of model decisions helps identify problematic model elements, enabling targeted data collection and modified training objectives to improve the model.

Test Users Test users, including professionals who select and administer tests, educators, administrators, and policymakers, depend on score reliability and interpretation validity to make

system-level or instructional decisions. Their reliance on the integrity and validity of scores to drive decisions is significant (AERA et al., 2014). Model explanations provide concrete evidence to validate the choice of the scoring model². This includes understanding whether the extracted features and scoring logic fully capture the rubric and the construct definition, and whether the internal structure of the automated scores align with the construct of interest (Bennett and Zhang, 2015).

2.2 The FGTI Interpretability Principles

We develop four foundational interpretability principles – Faithful, Grounded, Traceable, and Interchangeable (FGTI) – targeting the needs and benefits of large-scale assessment stakeholders from Section 2.1. Our first foundational principle is that explanations should be *faithful* (Jacovi and Goldberg, 2020). Faithfulness is an important requirement in many high-stakes applications of interpretable AI (Rudin, 2019). Similar expectations extend to assessments, and all of the needs and benefits outlined in Section 2.1 depend crucially on faithfulness.

Principle 1 (Faithful). *Explanations of scoring decisions should accurately reflect the computational mechanism behind the scoring model’s prediction.*

A notable example of **unfaithful** scoring explanations are texts produced by prompting LLMs to generate an explanation (e.g., Lee et al. (2024) and Li et al. (2025)). Stepwise reasoning verbalized by LLM through prompting strategies such as chain-of-thought (Wei et al., 2022) are not explanations of their internal computation (Sarkar, 2024) and often fail to reflect the model’s true reasoning behavior (Arcuschin et al., 2025, Turpin et al., 2023). Moreover, LLMs are highly sensitive to superficial changes in prompts and input text, frequently exhibiting inconsistent judgments (Wang et al., 2024). Therefore, even when LLMs achieve high scoring accuracy, prompting them to “explain their decisions” cannot reliably address the stakeholder needs identified in Section 2.1.

Next, the model should use meaningful features that are explicitly linked to each student’s work and rely only on those features for the downstream computation.

Principle 2 (Grounded). *Initial features computed by the scoring model should represent human-understandable, explicitly identifiable elements of student work and item task.*

Regardless of the routine used to derive those features, these feature values should possess meaning that is understandable to humans and be explicitly based both in the student work and item task. For instance, cosine similarity of sentence embeddings used as an input feature (e.g., Condor and Pardos (2024)) is less human-understandable than discrete features whose values are associated with clear, verbalizable meaning. Having features that are grounded addresses the need to scrutinize the features of the scoring engines.

How should the model process these features to ultimately produce a final score? Scoring is inherently an evidentiary reasoning process, where elements of the student response and item tasks serve as evidence to support the inference about student knowledge and skills that the score represents (DiCerbo et al., 2020, Mislevy, 2020). Stakeholders need to be able to inspect and interact with the internal structure of the scoring model to ensure soundness, construct-relevance, and fairness (Section 2.1). To meet this need, the model’s evidentiary reasoning process must be decomposable into clear, sequential steps that a human could reliably execute and possibly intervene. Our next 2 principles state that the scoring model should be conducive to this decomposition and intervention:

²More examples of validity arguments on the use of automated scoring can be found in (Bennett and Zhang, 2015, Table 7.7).

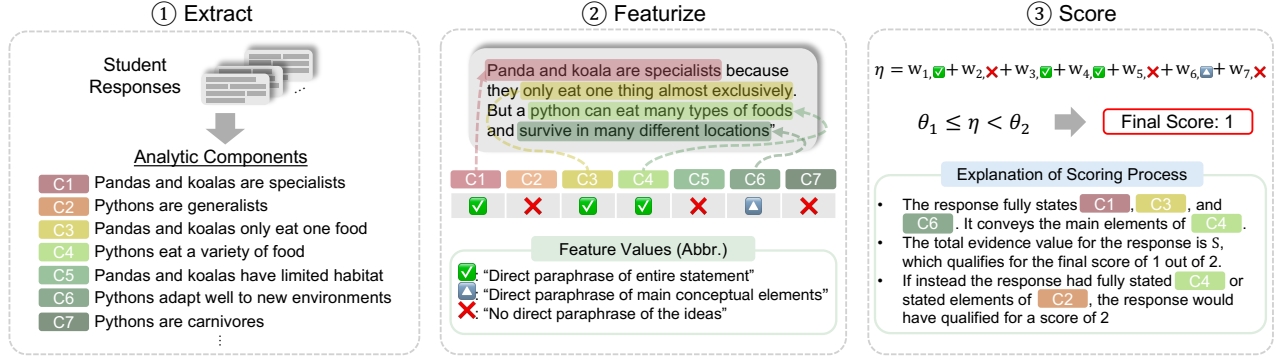


Figure 1: Schematic of the ANALYTICSCORE framework. The example question is: “Explain how pandas in China and koalas in Australia are similar, and how they both are different from pythons.”

Principle 3 (Traceable). *The scoring model should consist of subroutines that each represent a specific, well-defined evidentiary reasoning step on clearly specified inputs.*

Principle 4 (Interchangeable). *A human should be able to act interchangeably on each of the reasoning subroutines.*

Not all intermediate representations calculated by the model need to be understandable by human, but the reasoning subroutines should collectively account for the entire scoring logic. Moreover, humans should be able to act interchangeably with each subroutine and replace its outputs with human-generated results if deemed necessary.

Many proposed interpretability approaches are not grounded, traceable, or interchangeable in the sense described above. These include, for instance, calculating feature importance values (Asazuma et al., 2023, Kumar and Boulanger, 2020, 2021, Schlippe et al., 2022), displaying feature attribution maps (Li et al., 2025, Schlippe et al., 2022), or presenting confidence metrics for scoring decisions (Conijn et al., 2023). This limits the capacity to thoroughly inspect the model’s features and internal decision-making structure.

3 A Principled Framework for Interpretable Automated Scoring

To illustrate the feasibility of implementing the FGTI principles and to set a baseline for future research, we present ANALYTICSCORE as a reference framework in the domain of *short-answer scoring*. In this setting, students write a short 1-5 sentence answer in response to an assessment item which is scored with an emphasis on content correctness and demonstrated reasoning (Leacock and Chodorow, 2003, Shermis, 2015). The scoring model has access to a training set of student response texts paired with human-annotated scores $(r_1, s_1), \dots, (r_n, s_n)$ as well as unannotated responses $\{r_{n+1}, \dots, r_m\}$. The goal at inference time is to predict the score s for a new response r .

ANALYTICSCORE (Figure 1) is a 3-phase, LLM-based framework embodying the FGTI principles. Phase 1 identifies explicitly grounded *analytic components* to be used. Phase 2 catalogs, or *featurizes*, the presence of these components in student responses. Phase 3 uses the features to compute a score. Phases 1 and 2 depend only on the response texts without any annotations. Human score labels are only used during Phase 3. Source code implementing ANALYTICSCORE can be found at this link: <https://github.com/yunsungkim0908/analyticsscore>.

3.1 Phase 1: Extracting Analytic Components

With the response texts from the training set (and optionally the content of the assessment item), ANALYTICSORE first extracts a set of **analytic components**, which are explicitly identifiable elements of student responses described in Principle 2. For the short-answer scoring task considered in this work, we operationalize analytic components as propositions.

$$[c_1, \dots, c_k] = \text{Extract}(r_1, \dots, r_m),$$

Provided the extracted components align with the intended item design and are human understandable, any procedure can be used for component extraction. If necessary, these components could also be manually reviewed and modified by assessment designers. In this work, we implement component extraction by prompting an LLM with the prompts shown in Appendix A. Having too many analytic components could reduce interchangeability (Principle 4) by increasing the number of processed features (Lipton, 2018). We therefore limit extraction to 15 components per scoring unit.

3.2 Phase 2: Featurizing Responses

Once the analytic components have been identified, student responses are featurized according to the presence of these components $c_1, \dots, c_k$ in each response r . This step uses a labeling function $f(r; c)$ whose outputs are associated with human-understandable meaning (Principles 2 & 3). The exact label definitions used can be selected using natural language. In this work, we explore the following general purpose labeling function for $f(r; c)$. The precise definition of each score category can be found in Appendix A:

$$f(r; c) = \begin{cases} 2, & r \text{ contains direct paraphrase of } c \\ 1, & r \text{ contains partial paraphrase of } c \\ 0, & r \text{ doesn't contain paraphrase of } c \end{cases}$$

We implement $f(r; c)$ using Chain-of-Thought (Wei et al., 2022) prompting using the prompts shown in Appendix A. (Note that CoT is used solely as a prompting technique, and the generated “thoughts” are explicitly discarded.) Inspired by the self-consistency decoding strategy for LLMs (Wang et al., 2022), we apply the first-to-three aggregation rule to consider the possibly diverse interpretation of the labeling criteria when selecting the final output. Easily interpretable one-hot encodings of each $f(r; c)$ are then concatenated to produce a $3k$ -dimensional binary featurization of r :

$$F(r) = \text{OneHot}(f(r; c_1)) \parallel \dots \parallel \text{OneHot}(f(r; c_k))$$

Distilling LLM Featurizer into Open Source Using proprietary LLMs for featurization can quickly become too expensive in large-scale assessment settings, especially with many analytic components. To avoid the linearly growing cost of featurization, we supervised fine-tuned a small open-source model using a subsample of (r, c) pairs, where r is a response from the training set and c is an analytic component from Phase 1. More specifically, we randomly sampled 10k pairs across all 10 items, calculated the featurization labels on these samples using o4-mini, and collected the LLM model requests and outputs generated during this process that aligned with the aggregated final decision. This dataset was used to fine-tune Llama-3.1-8b-instruct with QLoRA (Dettmers et al., 2023).

3.3 Phase 3: Logically Traceable Scoring

Based on the featurized responses, a traceable and interchangeable model (Principle 3 and 4) is selected and trained using the labeled response pairs $(r_1, s_1), \dots, (r_n, s_n)$. Given the nature of the score

categories, we employ the Immediate-Threshold variant of Ordinal Logistic Regression (Pedregosa et al., 2017, Rennie and Srebro, 2005) as our scoring module. Combined with the one-hot encoding featurization from Phase 2, the resulting algorithm calculates the sum of weights for each component and feature label: $\eta = \sum_{i=1}^c w_{i,f(r,c_i)}$, where w are the trained weights. Scores are determined by comparing η to a set of learned thresholds θ_j ; the predicted score corresponds to the ordinal category j for which $\theta_j \leq \eta < \theta_{j+1}$. η can be understood as “evidence values” used for scoring.

3.4 Analysis of ANALYTICSCORE

An example of ANALYTICSCORE’s model explanation is shown in the far right panel of Figure 1. By demonstrating human-understandable features of the response (Principle 2) and the exact decision process (Principle 3), the explanation transparently and faithfully reveals the actual scoring mechanism used (Principle 1). If, based on the explanation, the model is suspected to have made an error (e.g., C6 should be a check, not a triangle), a human inspector can modify the featurization and recalculate the score by following the scoring algorithm (Principle 4), which is also how the “if instead...” explanation is generated.

The structure of ANALYTICSCORE’s scoring model is akin to Concept Bottleneck Models (Koh et al., 2020, Yang et al., 2023) in that we enforce a layer of intermediate representations with human-understandable “concepts.” Our framework ensures that the intermediate features have human-understandable values associated with explicitly identifiable elements of student work (Principle 2), as opposed to inferred characteristics.

4 Evaluating ANALYTICSCORE

We now evaluate ANALYTICSCORE’s scoring accuracy and how well its featurization aligns with human judgments on a real-world open-ended response dataset.

Dataset The ASAP-SAS dataset (Shermis, 2015)³ is the largest publicly available dataset of English short answer responses from US schoolchildren for 10 different open-ended exam questions, covering 3 areas of assessment: Science, Reading (Informational Text), and Reading (Literature Text). Human raters double-scored and assigned a single number to each student response using a 3 or 4 point rubric. The dataset details are reported in Appendix B. We use the original test set and split the public training set into training and validation sets with an 8:2 ratio.

ANALYTICSCORE Implementation Details For each assessment item, we used GPT-4 . 1 as the base LLM and extracted 15 analytic components except for Q7. This item uses a two-part scoring scheme to separately assess a character trait identified from the reading and its supporting evidence. We extracted 15 analytic components from each part, totaling 30 components. For the featurizer, we experimented with GPT-4.1-mini and Llama-3.1-8B-Instruct as our base LLM, each with temperature set to 0.7 and 1.0. Training and implementation details can be found in Appendix C.

4.1 Scoring Accuracy Experiment

We measured scoring accuracy in terms of quadratic weighted kappa (QWK) against the model scores in the test set, following the convention of the automated scoring literature (Institute of Education Statistics, 2023, Shermis, 2015).

³<https://www.kaggle.com/competitions/asap-sas/data>

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	All Avg.	Sci Avg.	R(Inf) Avg.	R(Lit) Avg.
Human Scorers	0.95	0.93	0.77	0.75	0.95	0.93	0.96	0.86	0.84	0.87	0.88±0.02	0.93±0.01	0.79±0.03	0.91±0.05
ANALYTICSCORE														
w/ GPT-4.1-mini*	0.80	0.86	0.64	0.59	0.79	0.78	0.61	0.59	0.80	0.68	0.72±0.03	0.78±0.03	0.68±0.06	0.60±0.01
w/ Llama-3.1-8b Inst* + Distillation*	0.57	0.57	0.59	0.56	0.69	0.47	0.52	0.45	0.74	0.60	0.58±0.03	0.58±0.04	0.63±0.05	0.48±0.04
	0.80	<u>0.82</u>	0.68	0.59	0.81	0.76	0.62	0.59	0.78	0.64	0.71±0.03	0.77±0.03	0.68±0.06	0.60±0.01
Few-Shot														
GPT-4.1	0.69	0.65	0.61	0.65	0.72	0.61	0.34	0.57	0.76	0.69	0.63±0.04	0.67±0.02	0.68±0.04	0.45±0.12
Supervised LLM														
BERT	0.80	0.80	<u>0.70</u>	0.70	0.80	0.81	0.69	0.68	0.84	0.71	0.75±0.02	0.79±0.02	0.74±0.05	0.69±0.01
DeBERTa	<u>0.85</u>	0.86	0.66	0.70	0.81	<u>0.83</u>	<u>0.71</u>	0.64	0.79	0.71	<u>0.76±0.03</u>	<u>0.81±0.03</u>	0.72±0.04	0.67±0.04
Llama-3.1-8b Inst.	0.84	0.73	0.72	<u>0.71</u>	<u>0.82</u>	0.81	<u>0.71</u>	<u>0.66</u>	<u>0.82</u>	<u>0.75</u>	<u>0.76±0.02</u>	0.79±0.02	<u>0.75±0.03</u>	0.69±0.02
w/ rubric	0.87	0.80	0.68	0.77	0.85	0.80	0.72	0.65	0.84	0.79	0.78±0.02	0.82±0.02	0.76±0.05	<u>0.68±0.04</u>
Llama-3.1-8b	0.83	0.75	<u>0.70</u>	0.77	<u>0.82</u>	0.84	0.68	0.65	<u>0.82</u>	0.74	<u>0.76±0.02</u>	0.80±0.02	0.76±0.03	0.67±0.02
Baseline														
AutoSAS	0.68	0.47	0.57	0.61	0.50	0.54	0.37	0.44	0.77	0.68	0.56±0.04	0.57±0.04	0.65±0.06	0.41±0.04
ASRRN	0.60	0.43	0.57	0.60	0.61	0.64	0.59	0.51	0.71	0.66	0.59±0.02	0.59±0.04	0.63±0.04	0.55±0.04
NAM*	0.63	0.62	0.43	0.35	0.72	0.63	0.42	0.38	0.76	0.62	0.56±0.05	0.64±0.02	0.52±0.13	0.40±0.02

Table 1: Test-time Quadratic Weighted Kappa (QWK) of scoring models per item, along with average per assessment area. **Best**, second-best, and *at human-level* performance scores are marked respectively. **Sci.**: Science (Q1,2,5,6). **R(Inf)**: Reading(Informational Text) (Q3,4,9). **R(Lit)**: Reading(Literature) (Q7,8). (*) are methods that are considered interpretable. Human scoring accuracy was taken from (Shermis, 2015).

We compare against the following baseline methods (See Appendix C for the list of hyperparameters used):

Few-Shot Prompting: We few-shot prompt **GPT-4.1** with 10 randomly selected responses from each score category, including a rubric for the score categories.

Supervised Fine-tuned LLM: The following LLM-based classifiers were fine-tuned on the response-score pairs: **BERT** (Devlin et al., 2019), **DeBERTa** (He et al., 2020), **Llama-3.1-8b**, and **Llama-3.1-8b-Instruct** (Grattafiori et al., 2024). We also fine-tune **Llama-3.1-8B-Instruct** with the scoring rubric added as a part of the input.

Automated Scorer Baselines: **AutoSAS** (Kumar et al., 2019), **AsRRN** (Li et al., 2023), and **NAM** (Condor and Pardos, 2024).

The only baseline method that has aspects of interpretability is NAM. This method requires hand-crafting a specific form of rubric describing the key phrases and concepts to be used by the response. Using sentence embeddings with n-gram matching as its features, this method implements a logistic regression score classifier. To implement this baseline, we replace the rubrics with the analytic components extracted by our ANALYTICSCORE.

4.2 Featurization Alignment Experiment

The feature labeling task described in Figure 2 was designed to produce human-understandable features (Principle 2). But how well does the LLM’s featurization behavior align with how humans

actually understand this task? Even more fundamentally, how well do humans themselves agree in their understanding of this task?

To answer these questions, we sampled 50 (response, analytic component) pairs for each of the 3 assessment areas. To ensure balanced representation, the sample included a balanced number of pairs from each of the three score categories, as initially determined by the GPT-4.1-mini featurizer. We then asked 7 human annotators to conduct the labeling task on these samples. The human annotators consisted of five volunteers from [ANONYMOUS] and two of the study’s authors. None of the annotators had prior exposure to the LLM’s featurization outputs, preventing any potential bias. All annotators had advanced academic training (PhD-level) and teaching experience, five of whom have been instructors at the primary, secondary, and/or post-secondary level. Additional details about the annotation process can be found in Appendix D.

Aggregate Human label was generated by majority voting (ties resolved randomly). We calculated inter-rater reliability among human labelers (Krippendorff’s α) and alignment between each LLM featurizer and aggregate human labels (QWK and class-wise F1). We report the 95% bootstrap CI of each metric, reweighting the sampling probability to account for the initial balanced sampling of score categories.

5 Experiment Results

5.1 Scoring Accuracy Results

Table 1 shows the results of the scoring accuracy experiments. Across items and within each assessment area, ANALYTICSCORE outperforms* several automated scoring baselines on average and, given its interpretability, achieves reasonable performance compared to state-of-the-art black-box models. Except for the untuned Llama featurizer, each ANALYTICSCORE variant outperforms* the few-shot prompting and automated scoring baselines. Compared to the best-performing models in each assessment area, ANALYTICSCORE is, on average, within 0.06 QWK for all items, 0.04 QWK for Science, 0.08 QWK for Reading (Informational Text), and 0.09 QWK for Reading (Literature) items. These results are noteworthy given that the FGTI principles strictly limit architectural flexibilities that often improve performance, such as subroutines that cannot be mapped to discrete reasoning steps or scoring components that are only partially articulable.

Also noticeable is the striking improvement* in the performance of the Llama featurizer post-distillation, with an average increase of 0.13 QWK. The distilled Llama featurizer performs comparably to both variants of GPT-4.1 mini. Increase in average QWK is most notable for Science items (+0.19), followed by Reading (Literature) (+0.12) and Reading (Informational Text) (+0.05).

5.2 Featurization Alignment Results

Table 3 displays the Krippendorff’s α measured among the human raters in conducting the featurization task from Section 3.2. For all assessment areas, we observe $0.667 \leq \alpha < 0.8$, which fall into a range of acceptable inter-rater reliability (Krippendorff, 2018). We interpret this as a good level of rater agreement on the featurization process defined in this work and acknowledge that there is still potential to refine and improve the task further.

Next, Table 2 shows alignment between models and majority human labels. Most notably, the distilled Llama featurizer achieves substantially high agreement with the aggregate human features

* $p < 0.05$ for Wilcoxon signed-rank test across all items. Due to small n , no area-specific difference was statistically significant.

Assessment Area	Featurizer Model	QWK	Label Distribution ⁴			Label-wise F1		
			2	1	0	2	1	0
Science	Human		15.32%	3.70%	80.98%			
	GPT-4.1-mini	(0.89, 0.89)	7.56%	12.59%	79.85%	(0.83, 0.84)	(0.20, 0.21)	(0.96, 0.96)
	o4-mini	(0.94, 0.95)	14.35%	8.17%	77.47%	(0.93, 0.93)	(0.49, 0.51)	(0.98, 0.98)
	Llama-3.1-8B (Distilled)	(0.90, 0.90)	11.54%	5.27%	83.19%	(0.89, 0.89)	(0.20, 0.22)	(0.97, 0.97)
Reading (Informational Text)	Human		11.48%	18.44%	70.08%			
	GPT-4.1-mini	(0.71, 0.72)	9.70%	22.01%	68.29%	(0.68, 0.69)	(0.54, 0.55)	(0.87, 0.88)
	o4-mini	(0.81, 0.82)	18.05%	16.54%	65.42%	(0.73, 0.74)	(0.69, 0.70)	(0.94, 0.94)
	Llama-3.1-8B (Distilled)	(0.72, 0.73)	20.20%	9.98%	69.82%	(0.61, 0.62)	(0.24, 0.26)	(0.91, 0.91)
Reading (Literature)	Human		9.33%	5.12%	85.55%			
	GPT-4.1-mini	(0.52, 0.54)	2.64%	10.67%	86.69%	(0.44, 0.46)	(0.67, 0.69)	(0.95, 0.95)
	o4-mini	(0.49, 0.51)	6.16%	6.64%	87.20%	(0.49, 0.51)	(0.10, 0.12)	(0.92, 0.92)
	Llama-3.1-8B (Distilled)	(0.81, 0.82)	7.45%	6.39%	86.16%	(0.86, 0.87)	(0.17, 0.19)	(0.92, 0.92)

Table 2: Alignment between LLM featurizers and Aggregate Human featurization obtained by majority voting for different models and assessment area. QWK and F1 values presented are 95% Bootstrap CI.

Assessment Area	Krippendorff’s α
Science	(0.718, 0.723)
Reading (Informational Text)	(0.696, 0.700)
Reading (Literature)	(0.672, 0.680)

Table 3: Inter-rater reliability among annotators for the featurization alignment experiment (95% bootstrap CI).

across all assessment areas. Other featurizers also achieve high agreement in Science and Reading (Information Text) but achieve moderate agreement in Reading (Literature).

F1 scores and label distribution for each feature label⁴ provide a more detailed insight and reveal areas for further improvement. Notice that the F1 score is exceptionally high (near or above 0.9) for label 0, and moderate-to-high (0.6~0.93) for label 2, with higher agreement for Science items. Yet, alignment for label 1 is moderate-to-low, ranging from 0.68 down to 0.11. We attribute this result to the relatively ambiguous nature of the label category 1, coupled with the rarity of label 1 in human rating. While LLM featurizers achieve high overall alignment with aggregate human featurization, future work should reduce ambiguity in the featurization task and ensure model predictions better reflect the distribution of human featurization behavior.

6 Discussion

6.1 Limitations and Future Work

Capturing Complex Scoring Logic While the average performance gap of 0.06 QWK between ANALYTICSCORE and the uninterpretable SOTA is meaningful given the architectural constraints

⁴For Aggregate Human and o4-mini, we applied prevalence weighting to the 50 study samples to estimate the label distribution across all (r, c) pairs.

imposed by the FGTI principles, there is still room to improve. One potential area of improvement is the choice of the scoring module used during Phase 3 (Section 3.3). Our demonstration of ANALYTICSCORE uses ordinal logistic regression as the traceable and interchangeable scoring module. While this implementation provides a strong baseline, it may not sufficiently capture the logical nuances required for scoring complex items. Future work should explore more complex yet traceable and interchangeable alternatives to ordinal logistic regression. Examples of possible alternatives are LLM workflows or agents which consist of modules that each compute a specific evidentiary reasoning step and human-understandable intermediate outputs.

Limited Benchmark Datasets The ASAP-SAS dataset is the largest publicly available dataset of complex open-ended short-answer scoring, but it only reflects a tiny fraction of assessment items, response types, domains, test-taker population, and assessment language. For empirical results to robustly generalize, automated scoring methods should be comprehensively evaluated on various datasets collected across different assessment settings. Given the limited amount of publicly available large-scale benchmarks for complex open-ended short-answer scoring, curation of additional datasets is necessary to support a more rigorous study of automated scoring.

Real-World Validation Our analysis of assessment stakeholder needs and interpretability principles was scoped to a theoretical study derived from the educational assessment literature. To identify and address the gaps that exist between theory and practice, field studies should be conducted in the future to validate the needs of the real-world assessment stakeholders and iteratively refine the design of interpretable scoring systems.

Measuring Alignment Our featurization alignment study (Sections 4.2 and 5.2) measures the model-to-human and human-to-human alignment in featurization behaviors. In addition to measuring alignment in featurization behaviors, it is also important to measure how the extracted analytic components and the featurization task align with the target constructs that the assessment item is intended to capture (Bejar et al., 2016). Analyzing construct alignment requires the knowledge of the design decisions involved in the development of the assessment items, which we did not have access to in our current study. We hope to see larger-scale studies on both featurization alignment and construct alignment in authentic assessment environments in the future.

6.2 Ethical Considerations

Student Privacy Our demonstration of ANALYTICSCORE involves the use of proprietary LLMs for extracting analytic components, featurizing responses, and creating the training dataset to distill the featurizer. When handling real-world assessment responses, care should be taken to avoid leakage of potentially sensitive or personally identifiable information contained in student responses.

Responsible Use of Model Interpretations To ensure that the assessment outcomes are valid, reliable, and fair, the use of automated scoring should be accompanied by the analysis of multiple sources of validity evidence (Bejar et al., 2016, Bennett and Bejar, 1998, Bennett and Zhang, 2015, Williamson et al., 2012). These sources include the features used by the scoring model, the model’s alignment with human judgments, how the model handles unusual or unexpected responses, and the consistency of scoring behavior across student populations (Bennett and Zhang, 2015).

Model interpretations are a means to an end for enabling these analyses, but having an interpretable scoring system does not, on its own, ensure that the assessment outcomes will be valid, reliable, and fair. Without a thoughtfully designed operational practice around the use of model interpretations,

these systems have the risk of creating a *false sense* of validity and trustworthiness when no proper auditing or mitigation is actually taking place. Beyond the technical and architectural aspects of interpretable automated scoring discussed in this work, it is also important to develop rigorous workflows, protocols, and systematic practices for using model interpretations to ensure that automated scoring systems are responsibly audited, monitored, and improved over time.

7 Conclusion

Despite a pressing need, the AI and education research community has yet to develop a practical interpretability framework for automated scoring in large-scale educational assessments. In this work, we presented a principled approach to address this challenge. We analyzed the needs and potential benefits of assessment stakeholders and developed four foundational principles of interpretable automated scoring. As a baseline framework for future research, we developed ANALYTICSCORE for short-answer scoring, which achieves promising scoring accuracy and demonstrates featurization behaviors that align with human judgment. We hope this work illuminates exciting new directions in developing practical and effective interpretable automated scoring methods for large-scale educational assessments.

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A Prompts

Figure 2 shows the exact prompts used to implement the feature labeling function from Section 3.2.

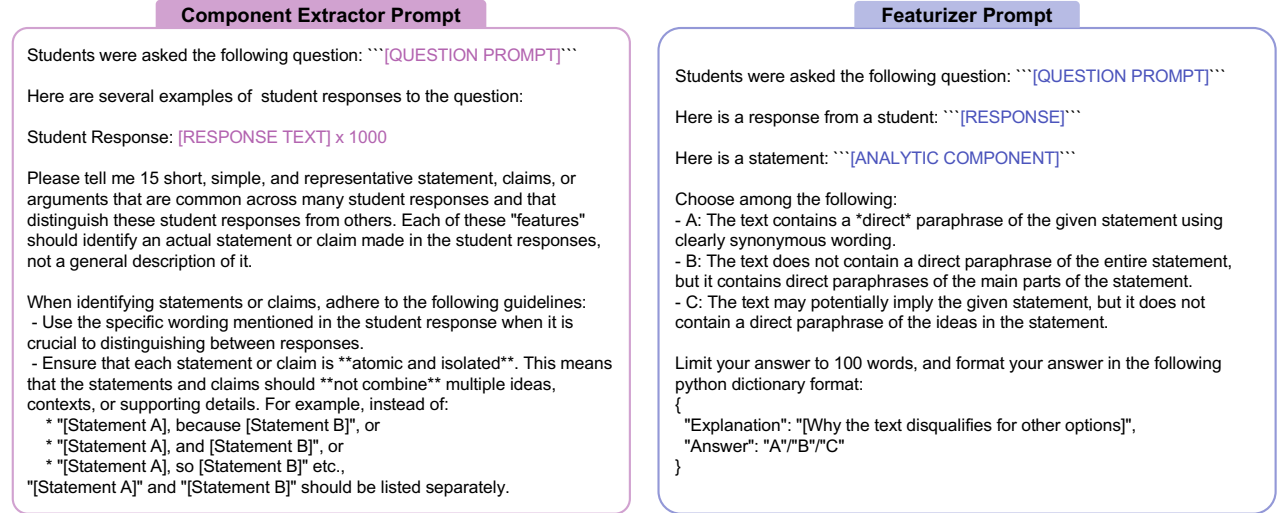


Figure 2: Prompts used in ANALYTICSCORE

B ASAP-SAS Dataset Detail

Table 4 shows the details of the ASAP-SAS dataset.

	Token Len.	Train	Valid	Test	Assessment Area
Q1	47.5 ± 22.2	1,341	331	557	Science
Q2	59.2 ± 22.6	1,024	254	426	
Q3	47.9 ± 14.6	1,445	363	406	Reading (Informational Text)
Q4	40.3 ± 15.5	1,308	349	295	
Q5	25.1 ± 21.5	1,459	336	598	Science
Q6	23.8 ± 22.6	1,418	379	599	
Q7	41.3 ± 25.1	1,432	367	599	Reading (Literature)
Q8	53.0 ± 32.6	1,446	353	599	
Q9	49.7 ± 36.3	1,453	345	599	Reading (Informational Text)
Q10	41.1 ± 28.5	1,314	326	546	Science

Table 4: ASAP-SAS dataset detail by item.

C Implementation Details

ANALYTICSCORE Implementation Details We distilled the Llama featurizer for 2 epochs using a batch size of 4 and learning rate of 1e-4. All model calls were made through the official OpenAI API. Fine-tuning was conducted on an Ubuntu 20.04 machine with 2 RTX A6000 GPUs (49Gb memory), 16 AMD EPYC 9224 24-Core Processors, and 250Gb of CPU RAM.

Baseline Implementation Details Below is the list of hyperparameters used in this work. Lists indicate the range of hyperparameters searched.

- **Featurizer Distillation**
Seed: 42; **Learning Rate:** 1e-4; **Epochs:** 2; **Batch Size:** 8; **Weight Decay:** 0.01; **Quantization:** 4-bit nf4; **LoRA Rank:** 16; **LoRA α :** 8; **LoRA Dropout:** 0.05
- **Few-Shot LLM**
Number of Fewshot examples: 10 random few-shot examples per score category, sampled for each response; **Temperature:** 0.7
- **SFT LLM (Encoder-Only Models)**
Seed: 42; **Learning rate:** 3e-5; **Batch Size:** 12; **Early Stopping:** Threshold 1e-3, patience 10; **Classifier Head:** 2-layer feedforward network with hidden dim 32, dropout probability 0.1
- **SFT LLM (Decoder-Only Models)**
Seed: 42; **Learning Rate:** 1e-4; **Batch Size:** 8; **Epochs:** 20; **Early Stopping:** Threshold 1e-3, Patience 5; **Weight Decay:** 0.01; **Quantization:** 8-bit; **LoRA Rank:** 16; **LoRA α :** 8; **LoRA Dropout:** 0.05
- **AutoSAS**
Maximum Depth: [50, 75, 100, 150, 200]; **Number of Estimators:** [50, 75, 100, 150, 200];
Hyperparameter selection criteria: Best QWK on validation set
Final selected hyperparameters:
 - Q1. Max Depth: 200; # of Estimators: 150
 - Q2. Max Depth: 200; # of Estimators: 75
 - Q3. Max Depth: 75; # of Estimators: 100
 - Q4. Max Depth: 75; # of Estimators: 50
 - Q5. Max Depth: 150; # of Estimators: 100
 - Q6. Max Depth: 50; # of Estimators: 150
 - Q7. Max Depth: 200; # of Estimators: 150
 - Q8. Max Depth: 200; # of Estimators: 150
 - Q9. Max Depth: 150; # of Estimators: 200
 - Q10. Max Depth: 200; # of Estimators: 50
- **AsRRN**
Parameters were mostly adopted from the following public implementation: <https://github.com/nkzhlee/AsRRN>
Seed: 23; **Batch size:** 1; **Learning rate:** 1e-5; **Epochs:** 23; **LR gamma:** 0.1; **LR step:** 15; **Max sequence length:** 128; **Validation split:** 0.2; **Weight decay:** 0.0001; **Max gradient norm:** 1.0; **Warmup steps:** 0.2; **Gradient accumulation steps:** 1; **Pretraining steps:** 5; **Hidden dimension:** 768; **Message dimension:** 128; **Hidden dropout probability:** 0.2; **Number of graph steps:** 2; **Contrastive loss lambda:** 0.01; **Contrastive temperature:** 1; **Norm:** 1

- **NAM N-gram sizes:** 2~10-grams

D Alignment Study Details

The annotators received an oral presentation of the purpose of the study along with links to 3 Qualtrics forms to be filled out, one in each assessment area. The form reiterated the study’s purpose, explained the task, and presented 50 items to annotate, each containing the context of the assessment item and the same featurizer prompt shown in Figure 2. The overall process took each annotator between 2.5 to 3.5 hours. Figures 3 through 5 are example screenshots from one of the three qualtrics forms used for annotation.

Welcome to our study!

This study is designed to evaluate a core component of a new, explainable AI autoscoring system we developed for open-ended student responses. This new system works by first identifying whether certain statements are explicitly stated in a student response. The goal of this study is to evaluate this component of the system by comparing our system's decisions to human judgments. Your responses will help us understand how closely the AI systems's detection behavior aligns with that of humans.

This study consists of **3 parts** containing a total of **50 items**. Each part corresponds to a distinct **assignment task** that students were asked to complete. At the top of each page, you will find a brief description of the original assignment task for that part. Each item presents:

- a **statement** representing a specific idea or claim, and
- a **student-generated response** for the assignment.

Your task is to assess, in the context of the assignment instruction, how directly the given statement is stated in the student response. Specifically, you'll be asked to choose one of the following choices:

- The response text contains a **direct** paraphrase of the given statement using clearly synonymous wording.
- The response text does not contain a direct paraphrase of the entire statement, but it contains direct paraphrases of the main conceptual elements of the statement.
- The response text may potentially imply the given statement, but it **does not** contain a direct paraphrase of the ideas in the statement.

Please **carefully** read each of the choices and choose the one that best applies to the given statement and student response text.

Keep in mind that the task is **NOT to judge the quality or correctness of the student response**, but instead to evaluate how directly the student response states the given statement in the context of the assignment instruction. Both the statement and the student response may be incorrect, but the response may still contain a direct paraphrase of the statement.

Thank you for participating in this study. Your judgments will play an important role in helping us evaluate and improve the explainability and interpretability of AI-based educational tools. Please proceed to the next page to begin the task.

Figure 3: Welcome page of one of the 3 Qualtrics forms used for the featurization alignment experiment. This page contains explanations about the purpose of the study, the necessary contexts, and task description.

Part A (18 items)

Assignment Description

In this **language arts** problem, students were provided an article exploring the controversy around rising exotic pet trade in the U.S., focusing on reptiles like pythons. The article discusses the conflict between traders who see them as harmless and biologists who warn they threaten native ecosystems as invasive species.

Part of the article makes a distinction between *generalist species* such as pythons that can adapt to wide range of environments, habitats, and diets, and *specialist species* such as pandas and koalas that depend on limited food source, have narrow habitat requirements, and favor stability over change.

The final instruction reads:

“ Explain how pandas in China are similar to koalas in Australia and how they both are different from pythons. Support your response with information from the article. ”

Figure 4: Example description of the assessment item and assessment instruction shown to the annotators.

#1. Please carefully read the choices and choose the one that best applies to the following statement and student response text in the context of the assignment instruction. .

Statement: "Pythons are generalists."

Student Response Text: Panda's in China are similar to koalas in Australia because they are both specialist's. In China, the panda eats almost nothing but bamboo and in Australia, the koala bear eats almost nothing but eucalyptus leaves. They're both different from pythons because other species are specialists, then pythons. Pythons are the generalists.

The response text contains a **direct** paraphrase of the given statement using clearly synonymous wording.

The response text does not contain a direct paraphrase of the entire statement, but it contains direct paraphrases of the main conceptual elements of the statement.

The response text may potentially imply the given statement, but it **does not** contain a direct paraphrase of the ideas in the statement.

Figure 5: Each annotation task presented annotators with a (response, analytic component) pair and asked them to select one of the three label options used in ANALYTICSCORE's labeling function.